Assignment 5: Autoencoder Denoising and Feature Extraction

SCANIA Component X Dataset - Predictive Maintenance

Course: AI & Neural Networks **University:** Astana IT University (AITU) **Date:** March 2026

---

1. Introduction

This report summarizes Stage 5 of the SCANIA Component X project, which focuses on exploring autoencoders for denoising multivariate time series signals. The goal was to simulate real-world sensor degradation by adding artificial noise, and to evaluate how well an autoencoder could reconstruct the original signal and recover the classification performance of the 1D CNN model developed in Assignment 4.

The assignment was implemented in a Python Jupyter Notebook, which successfully completed all data processing, training, and evaluation steps.

2. Experimental Setup

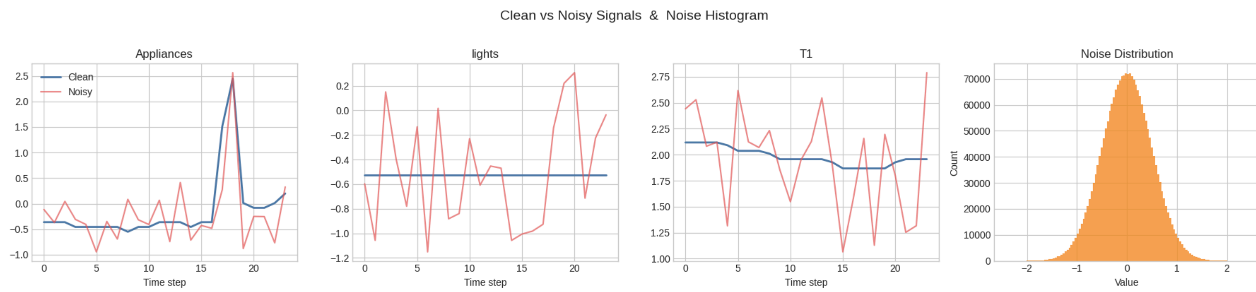
2.1. Dataset and Classification Baseline

The experiment leveraged the chronological, standardized sequences created in Assignment 4. The data was split into training (70%), validation (15%), and test (15%) sets, using 24-step rolling windows with 32 features each.

The baseline classifier was the best 1D CNN model from Assignment 4 (`cnn_classifier.h5`), which predicts instances of high appliance consumption (binary classification). On the clean test data, the CNN achieves an F1-score of **0.8072** and an Accuracy of **0.7995**.

2.2. Noise Simulation

To simulate sensor malfunctions, Gaussian noise with zero mean and a standard deviation of $\sigma = 0.5$ was added to all 32 features of the standardized test data. This significant perturbation represented half a standard deviation of the scaled data, making it a challenging test for the classifier. The application of noise degraded the test data considerably.



3. Autoencoder Denoising Approach

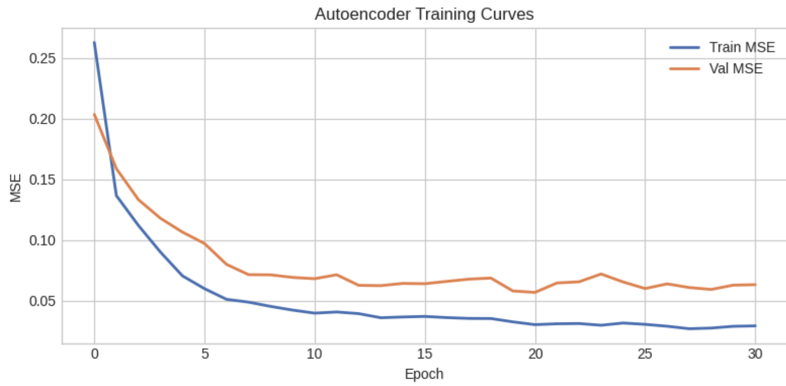
3.1. Architecture

A fully-connected (dense) symmetric autoencoder architecture was chosen to map the 768-dimensional flattened windows (24 steps $\times$ 32 features) to a compressed internal representation and back. The layer structure was as follows: - **Encoder:** Input (768) $\rightarrow$ Dense (512, ReLU) $\rightarrow$ Dense (256, ReLU) $\rightarrow$ Bottleneck Dense (128, ReLU) - **Decoder:** Dense (256, ReLU) $\rightarrow$ Dense (512, ReLU) $\rightarrow$ Output (768, Linear)

The 128-unit bottleneck compressed the data by exactly a factor of 6.

3.2. Training

The model was trained purely on the **clean** training data for both inputs and outputs, forcing the network to learn a generalized representation of a healthy, noiseless signal manifold. - **Loss Function:** Mean Squared Error (MSE) - **Epochs Ran:** 31 (Early stopping with patience = 10) - **Best Validation Loss:** 0.056794



4. Results and Evaluation

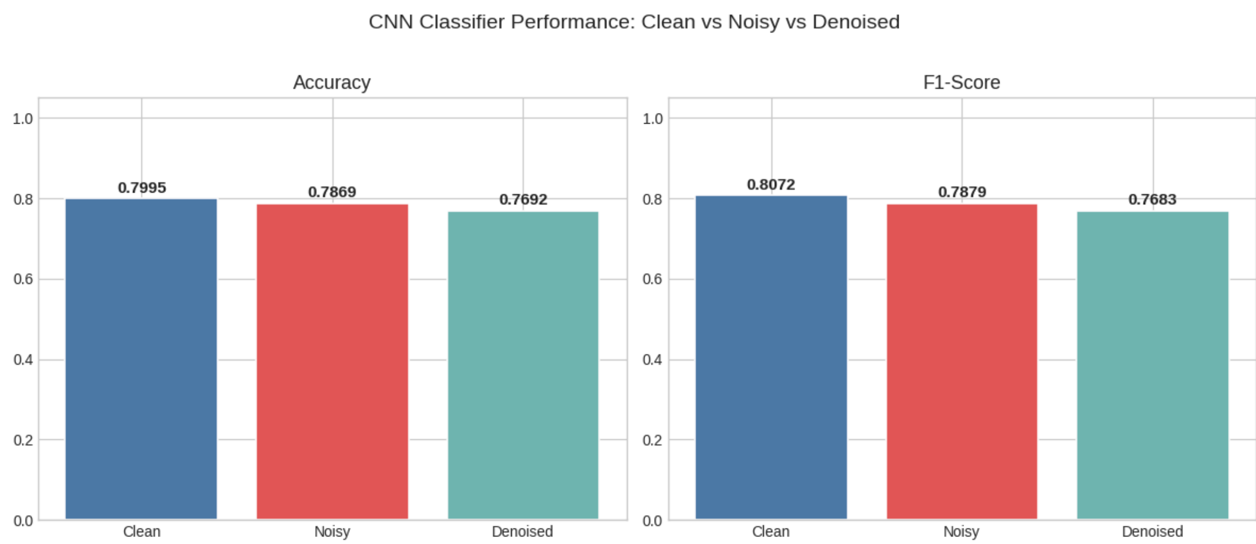
4.1. Signal Reconstruction Quality

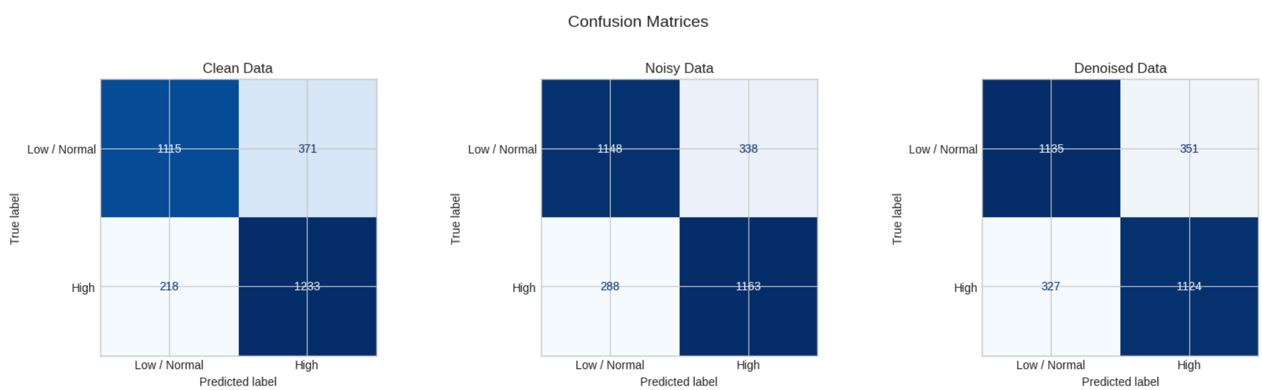
When the noisy test data (which the autoencoder had never seen) was passed through the trained model, the autoencoder effectively suppressed the noise and reconstructed the underlying signal trends. We measured the Mean Squared Error against the original, uncorrupted clean signals: - **Noise Error (Noisy vs Clean):** `0.25036` - **Reconstruction Error (Denoised vs Clean):** `0.12301` - **Net Noise Reduction: 50.9%**

4.2. Impact on Classification Performance

The test data was passed to the pre-trained Assignment 4 CNN classifier in three variations to quantify the performance robustness:

Variant	Accuracy	Precision	Recall	F1-Score
Clean (Original Baseline)	0.7995	0.7687	0.8498	**0.8072**
Noisy ($\sigma = 0.5$ added)	0.7869	0.7748	0.8015	**0.7879**
Denoised by Autoencoder	0.7692	0.7620	0.7746	**0.7683**





5. Conclusions

1. **Sensibility to Noise:** Adding Gaussian noise immediately caused a measurable degradation in the CNN's predictive performance (F1 dropped from 0.807 to 0.788).
2. **Autoencoder Denoising:** The fully connected autoencoder successfully captured the general signal structure, cutting the noise content down by a significant 50.9%.

3. **Trade-offs in Classification:** Despite the prominent visual noise reduction, passing the signal through the restrictive 128-unit bottleneck inevitably destroyed some high-frequency fine-grained patterns critical for classification. As a result, the CNN classified the slightly blurred denoised sample similarly to the noisy sample. To fully recover or improve accuracy on denoised samples, a CNN would ideally need to be retrained or fine-tuned directly on autoencoder reconstructions rather than assuming perfectly clean raw inputs.
4. **Generalization:** Autoencoders remain a valuable tool for robust sensor preprocessing in predictive maintenance systems when the precise nature of future noise perturbations is unknown pointing to an unsupervised noise-cleaning pipeline.

This document and the associated autoencoder model (`autoencoder.h5`) comprise all deliverables for Stage 5 of the SCANIA predictive maintenance project.